Short Laws, Discrete Steps, Forward Time: Why Gravity and Space Are Causal Transitions

Abstract

Everything that exists is mathematics, which answers three questions: why the laws of physics are short, why causes take discrete steps, and why time runs forward. Whatever exists is some definite way, each definite way is ruled or random, and both are mathematics. Anything that acts on the world, a soul, a designer or a simulator, acts by definite relations, so it is mathematics too. No probability distribution treats infinitely many laws alike, and the weighting that needs no extra numbers flips a fair coin for each bit and for whether to add another. Each bit halves a law's weight, and unless longer laws grow observers more often, the shortest laws that can grow them carry at least half the weight. The known laws fit on a page, as this predicts. A differential equation encodes dependence among states. Production takes a state to its next, and a continuum has no next state, so production needs discrete steps. Physics agrees: gravity and quantum theory suggest a finite region holds finitely many distinguishable states, reached at a finite rate, and the order of the steps, counted, gives spacetime its geometry. Short laws mean a simple start, and minds are complex, so the universe has to grow them in causal steps. An endless chain of steps has no last step to run back from, a finite reversal pays extra bits to name its turning step, about 200 for a typical one, and minds merge many states into one, so they run one way.

1. Four questions, one chain

Eugene Wigner called the usefulness of mathematics in physics unreasonable (Wigner, 1960). On the view taken here it is expected. Everything is ruled or random, both are mathematics, and whatever acts on the world acts by rules, so physics has nothing else to describe.

Three more questions then have answers a reader can check, and the answers form a chain. The laws of physics are short because each bit a description adds halves its probability. Short laws mean a simple start. Minds like ours are complex, so a simple start has to grow them, and growth takes causal steps, each state produced from the one before. The steps are discrete because a cause has to produce a next state, and a continuum has no next state. Alfred North Whitehead reached the same conclusion from Zeno's paradoxes (Whitehead, 1978). Time is the sequence of steps, and it runs forward, from a simple beginning toward a complex present. Light, which crosses space while no time passes for it, is no exception: a moving body's transitions run slower, and light spends every step moving, so nothing within it could move relative to the rest. An endless chain of steps has no last step to run back from, a reversal has to pay to name its turning step, and minds merge many states into one, so they run one way.

Each check takes a few lines. One concerns probability on a countable set, a set whose members can be listed: no probability distribution treats infinitely many laws alike, and a fair coin, flipped for each bit and for whether to add another, makes long laws lose. Another concerns continua, gapless sets such as the real line: a continuum has no next state, so causes take steps, an a priori proof in the logician's sense. The last is a count: fewer than 2^m programs are shorter than m bits, so most of N steps cost about $\log_2 N$ bits to name.

2. Why everything is mathematics

Whatever exists is some definite way, or nothing would tell it apart from nothing. Each definite way is ruled or random. Either a rule shorter than the structure generates it, or nothing shorter does. The length of the shortest program that prints a string is its Kolmogorov complexity, and a string that no shorter program prints is random in the precise sense of that theory (Li and Vitányi, 2019). Rules are mathematics. A random string is mathematics too, with definite entries in definite places. Every finite string falls on one side or the other, whether or not any proof can say which (Chaitin, 1974). No third kind remains for a nonmathematical thing to be.

Interaction gives the same answer from the other side. In 1643 Princess Elisabeth of Bohemia asked René Descartes how a soul that only thinks could move a body (Shapiro, 2007). Any answer has to say how the soul acts: what it changes, under which conditions and by how much. Those when-then relations are rules, and rules are mathematics. Whatever acts on the world, a soul, a designer or a simulator, acts by such rules, so it belongs to one mathematical structure with the world it acts on. A thing that acts on nothing explains nothing here.

The last refuge is an intrinsic nature behind the relations, what a thing is in itself. Swap the hidden natures behind mass and charge while every relation stays fixed, and nothing anyone could notice, think or say would change (Lewis, 2009). A nature that makes a difference takes part in relations, so it is structure. A nature that makes none does no work, and nobody could know it.

Wigner's puzzle then dissolves. Mathematics describes the world well because the world is mathematics (Tegmark, 2008). What remains is which mathematics an observer should expect to find.

3. No fair lottery among infinitely many laws

Write every candidate law of nature as a finite string of 0s and 1s: its complete description, with equations, constants and starting state together. There are infinitely many such strings, and they can be listed, shortest first. A natural wish is to give them all the same chance, and it cannot be granted. A probability distribution assigns each outcome a chance, and under the standard axioms the chances of all the outcomes, countably many here, add up to one. If each string had the same probability p, the sum over infinitely many would be infinite when p is positive and zero when p is zero. Any distribution has to fall as descriptions grow.

The bits of a shortest description say how fast. A shortest description cannot be compressed, or the compressed version would be shorter still, so no rule inside it predicts one bit from the others, and it is random in the sense of Section 2. With no rule favoring 0 or 1, each bit of a shortest description is a fair coin. A biased coin would carry its bias as a number, and that number would have to be written into the description too.

Each added bit therefore halves a description's weight: a law of L bits weighs 2^{-L}. Descriptions that end on their own, none the beginning of another, can take these weights together, since by Kraft's inequality their total is at most one (Li and Vitányi, 2019). Ray Solomonoff used the same weights to predict data (Solomonoff, 1964). Read as weights on what exists, they say which laws an observer should expect to find.

Counting gives the opposite verdict, and the coin shows why counting misleads. There are 2^n strings of length n, twice as many with each added bit. A catalog of possible laws is therefore dominated by long ones: endless special cases, arbitrary constants, exceptions written out over libraries. Whether a description goes on at all is a second yes-or-no that no rule decides, so it is a fair coin as well. Half of the descriptions still going stop at every step. All descriptions of length n

together then weigh $2^{-(n+1)}$, and everything longer than n, added up, weighs exactly as much as length n alone: the tail equals the peak.

In this picture one particular description loses more than half its weight per bit, since each bit also pays for its stop-or-go flip. A code that states the length once, in a short header, loses only about half (Li and Vitányi, 2019). Either way, long laws lose.

The known laws, the Standard Model of particle physics with general relativity and their constants, fit on a page. If the count of strings decided, we should expect laws of endless special cases. We find laws that fit on a page. Two explanations remain. Either the weight halves with each bit, or something chose our laws: a designer, or a civilization running our universe as a simulation (Bostrom, 2003). A chooser moves the question up one level. Its own world needs laws, those laws face the same weighting, and the account is postponed.

4. Short laws must grow their observers

The very shortest laws produce nothing worth observing, and we are here. The laws to expect are the shortest that can produce observers: structures that record and model their surroundings. Added rules more often break the conditions for observers than create them, as small changes to the constants of our physics often remove stable nuclei, long-lived stars or chemistry (Barnes, 2012).

Under the coin-flip weights, suppose no longer length holds a larger share of observer-producing descriptions than the shortest length that works. Then that length carries at least half of all the observer-producing weight, since everything beyond it weighs no more than it does. A mind assembled by a chance fluctuation, a Boltzmann brain, needs every one of its bits supplied by chance, so minds grown by short rules outweigh it.

Short laws therefore mean a simple start: the laws and the starting state together hold little information. Minds are complex. A simple start has to grow them, and simple rules can. John Conway's Game of Life is a grid of cells that live or die by counting their live neighbors. In it, gliders travel across the grid, guns fire streams of gliders, and patterns built from them compute anything a computer can (Berlekamp, Conway and Guy, 1982). Rule 110, a row of cells each updated from itself and its two neighbors by a table of eight cases, is universal as well (Cook, 2004). A rule that fits on a line can build complexity without limit, one step at a time, each state produced from the one before. Producing is one kind of cause.

5. Two kinds of cause

Causes come in two kinds, as Ned Hall showed (Hall, 2004). Dependence relates states: change an earlier one and a later one changes. Production brings a state about. A plant that dies because nobody watered it depends on the omission, yet not watering produced nothing; the plant's own chemistry, running without water, produced the death. Each frame of a film fits the physics of the frame before, yet no frame produces the next. The projector does.

A differential equation, an equation relating a quantity to its own rates of change, encodes dependence. Under $x = vt$ every position depends on the start and the speed. Bertrand Russell answered the demand for a next place by holding that motion is only being at different places at different times (Russell, 1903). Positions at every instant with nothing producing the next are the emperor's new points. Dependence of this kind sits comfortably in a continuum. Production does not, because a continuum has no next state: between any two instants lies a third.

Read a continuous law as a rule for producing states and it becomes a flow. A flow is a family of maps φ_t, one for every positive time t, each taking a state to the state a time t later, with $\varphi_{s+t} = \varphi_s \circ \varphi_t$: one map applied after the other. Every

transition the flow defines is two smaller ones composed: $\varphi_t = \varphi_{t/2} \circ \varphi_{t/2}$. A flow has no smallest transition, so producing any later state would take infinitely many productions before the first.

Zeno's runner meets the same regress: he must reach the halfway mark before the finish and the quarter mark before that, so no first stretch exists for him to run (Aristotle, 1984, Physics VI.9). Read the other way, the flow hands out every later state at once. It pairs times with states, as Russell's answer does, and the pairing exists whether or not anything happens, like a frozen record. Whitehead, arguing by Zeno's method, put it this way: once anything becomes, there is "no continuity of becoming" (Whitehead, 1978, p. 35).

A discrete rule has what the flow lacks. A map f that takes each state to the next, $x_{n+1} = f(x_n)$, has a first application and a next state after every state. We change: our states follow one another. In a world that is mathematics, rules take each state to the next one, and a rule that does so needs a next state, which only steps supply. A change that nothing produces would be a change without a cause, and nothing in an account that allowed it could say why it happens. Stated precisely, a process that adds finitely or countably many events at each natural-number stage produces at most countably many events, and no uncountable continuum of them. Nothing here depends on our universe. Wherever anything happens, it happens in steps.

6. What physics adds

A reader who takes dependence to be all of causation finds no contradiction in a continuum, and for that reader the case rests on physics. Two results point the same way.

First, gravity and quantum theory cap the states a region can hold. A black hole's entropy, read as a count of its states, is a quarter of its horizon's area measured

in squares of the Planck length, about 1.6×10^{-35} meters (Bekenstein, 1973; Hawking, 1975). No region can hold more states than the black hole it would become: if it could, collapsing it would lower the entropy, which the second law of thermodynamics forbids. A region's capacity is therefore capped by the area of its boundary ('t Hooft, 1993; Susskind, 1995; Bousso, 2002). A ball one centimeter in radius holds at most about 10^{66} bits.

The bound is proved only in limited settings, such as free fields in weak gravity (Bousso, Casini, Fisher and Maldacena, 2014), and most researchers in quantum gravity expect it to hold in general. If it holds, a finite region has finitely many distinguishable states (Bao, Carroll and Singh, 2017).

Second, change has a speed limit. A system with average energy E above its lowest state needs at least a time $h/(4E)$ to reach a state it can be told apart from, where h is Planck's constant (Margolus and Levitin, 1998). In a finite time it passes through finitely many distinguishable states. Finite states reached at a finite rate leave nothing observable that needs a continuum beneath it. Quantum amplitudes, the complex numbers that give the odds of outcomes, form a continuum too. A computation reaches any amplitude in finitely many steps, to whatever precision a measurement can check, so amplitudes too describe what the steps produce.

Einstein's equations are written smooth, and that does not make spacetime smooth. A formula can say how hard each step pushes, and which way, for any input. Numerical relativity, the computer solution of Einstein's equations, runs them one step at a time to any precision wanted. It predicted the gravitational waves of merging black holes that detectors later recorded (Pretorius, 2005). The formula is a rule for the steps.

Light moves at the base rate, one step of distance per step of time. In the Game of Life the same limit is one cell per generation, and a glider moves at a quarter of

it. Iwo Bialynicki-Birula wrote Maxwell's equations, the laws of light, as a quantum cellular automaton: a grid of cells, each updated from its neighbors at every tick. At long wavelengths it recovers the continuous equations (Bialynicki-Birula, 1994). Light also acts on matter in whole quanta: each photon is absorbed whole, in one event.

Light seems to cut time loose from the steps, since it crosses space step by step while no time passes for it. Its proper time, the time a clock carried along would record, is zero because along a light path every step goes into moving, and nothing changes in place. A slower body shares the base rate between moving and changing in place, so a moving clock runs slow. The shares add like the sides of a right triangle, $(c\,dt)^2 = dx^2 + (c\,d\tau)^2$, with τ the moving clock's own time: the interval of special relativity, restated rather than derived. A thing's time is its transitions in place, and within light nothing could move relative to the rest, so light is the limit of a moving clock running slow, not an exception to time being the steps.

The steps carry gravity's geometry with them. Take a spacetime without paths that loop into their own past. There the order of which events can influence which fixes the topology, how its regions connect, the smooth structure and the geometry up to a local scale (Hawking, King and McCarthy, 1976; Malament, 1977). Counting supplies the scale. Causal set theory builds spacetime from exactly these two ingredients, an order and a count (Bombelli, Lee, Meyer and Sorkin, 1987; Surya, 2019). Gravity, the curving of spacetime, is then the order of the steps, counted.

The count made a prediction. A random count fluctuates by about its square root, and Rafael Sorkin argued that the fluctuation leaves a cosmological constant, the energy of empty space, of about 10^{-120} in Planck units (Sorkin, 1997). Supernova surveys found a constant of that order the next year (Riess et al.,

1998; Perlmutter et al., 1999), where quantum field theory on a continuum overshoots by about 120 orders of magnitude.

Discreteness need not mean a grid. On a square grid the shortest route to a diagonal point stays longer than the straight line, however fine the grid becomes. Hermann Weyl pressed this obstruction (Weyl, 1949), and Tobias Fritz generalized it to repeating networks (Fritz, 2013). Observed Lorentz symmetry, the sameness of physics for every observer moving at a steady speed, rules out a regular lattice at the base. A random scattering of events, as in causal sets, picks out no direction (Bombelli, Henson and Sorkin, 2009). A rule that rewrites relations step by step leaves a causal order behind, the record of which update depended on which, so the search for the final rule can run among short rewriting rules, shortest first, keeping those whose records, at large scales, scatter like a causal set. It can also start from the known laws, written as a program of steps, cleared of their infinities and shortened while its outputs still match.

7. What stays continuous

Calculus stays. It describes many steps at once, as the gas laws describe a crowd of molecules without tracking any one. A Riemann sum adds value times width over small cells, and its limit describes a crowd of steps; the limit is the description, and the steps are what happen. Discreteness gives nothing a size. A discrete point has no width either, so in both pictures length comes from the relations among points. A grain with a width would bring the continuum back as its surface. What discreteness supplies is a next step, the one thing causation needs. Choice goes the same way. Analysis may pick a member from each of uncountably many sets with no rule for picking, the axiom of choice, and it then has sets of points with no volume and the doubling of a ball (Banach and Tarski, 1924). A history makes finitely many picks per step, countably many in all, and

with only such choices every set of reals can be measurable (Solovay, 1970). The doubling exists as mathematics, and no history performs it.

On the view taken here, every consistent structure exists as mathematics, continua included (Tegmark, 2008). What happens, in any of them, comes in steps, and the rules that take each step are the only things doing the work: rules applied to rules, with nothing outside the mathematics applying them.

8. Why time runs forward

Time is the sequence of causal steps. Short laws give a simple start, the start has to grow its observers, and growth runs from simple to complex. Gravity ties that simple start to the second law of thermodynamics, the rise of entropy. Without gravity an evenly spread gas has the most entropy. Under gravity, clumping raises it, so a smooth early universe, the simplest start to describe, is also a low-entropy one (Penrose, 2004). The early universe was even to about one part in 100,000 (Planck Collaboration, 2020).

Three facts keep the steps from running the other way. First, an endless chain of steps has a first step and no last one. Forward, state n+1 is produced from state n, beginning from a supplied first state, and a rule that is not told to stop goes on without end. Backward, state n would wait on state n+1, and with no last state the reversed chain has no first step that can run. The whole history has no end to run back from.

Second, a finite reversal has to name its turning step. A billiard break shows why. One stroke sends fifteen balls scattering, with sound through the room and warmth into the table. Reversing it takes every ball, air molecule and vibration in the table converging in perfect coordination, and a description written from the end has to list them all. A finished record costs about the same to describe in either direction: for a string h and its reversal h^R, $K(h^R) \leq K(h) + c$ for a constant c

that does not depend on h. The difference lies in production.

The short way to produce the reversed history runs the forward one, stops at a chosen step and turns around, and the step has to be named. Fourteen billion years hold about 8×10^{60} Planck times, each the time light takes to cross a Planck length, a number between 2^{202} and 2^{203}. A few special steps, such as those numbered by powers of two, have short names, though even a short name costs bits that the endless forward history does not pay. Most have none, since fewer than 2^m programs are shorter than m bits, so a typical turning step costs about 200 bits (Li and Vitányi, 2019). Under the coin-flip weights that costs a factor of 10^{-60} or less.

Third, minds merge states. Adding 1 and 1 gives 2, and so does adding 0 and 2. A brain's step takes many possible earlier states to one later state, and the difference leaves as heat (Landauer, 1961). Forward, a brain's present state and input fix its next state. Backward, $2 = x + y$ has more than one solution, and nothing in the brain picks one. In a reversed history heat flowing in from the surroundings in perfect coordination would pick it, and a brain whose next state is set from outside is not producing its own thoughts. Memories are built by steps already taken, and a memory that need not be fine-tuned to what it records generically points the way entropy rises (Mlodinow and Brun, 2014). We remember the past and not the future.

9. Conclusion

Everything is ruled or random, and both are mathematics, so whatever acts on the world, a soul, a designer or a simulator, is mathematics too. The laws are short because each bit halves a description's weight. No distribution can treat infinitely many laws alike, and the fair coin, flipped for each bit and for whether to add another, is the weighting that needs nothing written down beyond the law itself.

Under it, unless longer laws grow observers more often, the shortest laws that can grow them carry at least half the weight. Everything those laws do happens in steps, because a cause produces a next state, and a continuum has no next state.

Physics agrees wherever it reaches: finitely many states in a region if the entropy bound holds, a finite rate of change, smooth equations serving as rules for steps, light at the base rate, and geometry from the order of the steps, counted. Gravity and space are those causal transitions, and time is their sequence, with transitions within moving frames running slower, down to none for anything at light speed, such as a photon. Time runs forward because a simple start must grow its observers in causal steps. The causal order is a priori; the simple start comes from the weighting. An endless chain has no last step to run back from, a reversal pays extra bits to name its turning step, about 200 for a typical one, and minds merge many states into one. The continuum keeps its place in mathematics, as the description of many steps at once.

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